Cenwei Zhang

(+86)182-4007-8610 | zcw12138@sjtu.edu.cn | blog.cenweizhang.github.io | github.com/cenweizhang

Educations

The Chinese University of Hong Kong (CUHK) | Information Engineering *Postgraduate*

Sep. 2026 (Expected) — Expected Graduation: 2028

Shanghai Jiao Tong University (SJTU) | Biomedical Engineering (EE Track) *Undergraduate*

Sep. 2022 — Jun. 2026 (Expected)

GPA: 3.63/4.3. Awarded Academic Scholarships multiple times; Second Prize in the National Undergraduate Biomedical Engineering Innovation and Entrepreneurship Competition.

Graduation Thesis: **A Hierarchical Distribution-Preserving Pruning Framework for SAM on Medical Images, combining geometry-aware structural pruning and progressive fine-grained sparsification for the efficient deployment of foundation segmentation models**.

Skills

- **Languages**: Python, C++; proficient in the PyTorch deep learning framework.
- **Workflow**: Linux, Shell, Git, GitHub.
- **Technical Foundation**: Solid understanding of **Generative Modeling** theories (Flow Matching, Diffusion, AR) and on-manifold optimal transport mechanisms; proficient in backbone architectures including **Transformer, DiT, and ViT**; familiar with foundational LLM frameworks (via nanoGPT reproduction and fine-tuning); knowledgeable in Embodied AI and VLA models (e.g., pi0, RDT).
- **Others**: Hands-on experience with containerization technologies; proficient in Docker.

Work Experience

DolphinAI Co., Ltd. |*Algorithm Research Intern | Medical Image Generation* Apr. 2025 — Dec. 2025

- **Generative Model Research and Validation**: Independently spearheaded liver ultrasound and multimodal medical image generation projects using **Diffusion Models**. Addressed domain-specific challenges such as high-frequency noise and fine-grained feature preservation, conducting comprehensive feasibility studies from algorithm selection to generation quality assessment.
- **Algorithm Engineering and Containerization**: Bridged the gap between theoretical research and industrial deployment. Proficiently utilized **Docker** containerization to establish standardized environments for model training and inference, effectively resolving cross-platform dependency bottlenecks.
- **Synthetic Data Pipeline Construction**: Pioneered and consolidated a synthetic data generation pipeline tailored for medical scenarios. Delivered robust algorithmic baselines and scalable engineering workflows for data augmentation, directly empowering downstream disease diagnosis models.

Selected Publications

Axiomatic On-Manifold Shapley via Optimal Generative Flows |***Under Review at ICML*** *(CCF-A)*

Cenwei Zhang*, *Lin Zhu**, *Manxi Lin, Lei You*†

- **Generative Flows & Optimal Transport Foundation**: Formulated a continuous geometric framework based on optimal generative flows. Utilized kinetic-energy-minimizing Wasserstein-2 geodesics to define exact transport paths on high-dimensional data manifolds, effectively resolving path ambiguity and discretization drift.
- **Axiomatic Model Attribution System**: Derived and established the gradient line integral as the unique functional satisfying geometric efficiency axioms. Fundamentally eliminated arbitrary baseline selections and "off-manifold artifacts" in interpreting deep generative models and complex neural networks.
- **Quantitative Evaluation & Theoretical Guarantees**: Proposed alignment evaluation metrics, including Flow Consistency Error (FCE), outperforming existing SOTA methods across multiple generative benchmarks. Provided provable mathematical stability bounds to robustly handle ODE/SDE approximation errors.

GraphGaitNet: Anatomically-Informed Graph Neural Networks on Ground Reaction Force for Parkinson's Disease Diagnosis |***Accepted to IEEE JBHI*** *(CCF-B)*

Wenhao Li, ***Cenwei Zhang****, Shi Chang, Jingtong Zhao, Guanning Lin*†

- **Graph Representation for Gait Analysis**: Contributed to the design of the GraphGaitNet framework. Constructed Ground Reaction Force (GRF) graph representations based on plantar anatomical structures and force transmission patterns to capture local and global spatial dependencies for Parkinson's disease diagnosis.
- **Spatiotemporal Architecture & Evaluation**: Co-developed and evaluated the multi-layer residual graph convolution and Transformer-based temporal encoding modules. Validated the model on the PhysioNet Parkinson's gait dataset, achieving a state-of-the-art diagnostic accuracy of **96.8%**.

Projects

LLM Pre-training and Fine-tuning System based on nanoGPT |*Independent Project & Implementation*

Jan. 2026 — Mar. 2026

- **Model Architecture Reproduction**: Engineered the GPT-2 training and inference pipelines from scratch using PyTorch. Successfully implemented core modules including Multi-head Self-Attention, Causal Masking, and Autoregressive (AR) generation logic.
- **Architecture & Operator Optimization**: Upgraded the foundational framework with modern LLM designs such as Rotary Position Embedding (RoPE) and RMSNorm. Integrated the FlashAttention operator to significantly optimize memory footprint and computational efficiency in long-sequence modeling.
- **Pre-training Pipeline Construction**: Built an end-to-end pre-training and validation pipeline on a single H100 80G GPU. Employed `bfloat16` mixed precision and gradient accumulation strategies to maximize training throughput and hardware utilization efficiency.

Gait Analysis System based on Graph Neural Networks and Temporal Modeling |*Research Collaboration*

Oct. 2024 — Apr. 2025

- Co-designed and implemented the GraphGaitNet joint model for gait analysis, leveraging Ground Reaction Force (GRF) graph structures and human anatomical priors.
- Contributed to the engineering and experimental evaluation of the multi-layer residual graph convolution and temporal modeling modules.
- Modeled spatiotemporal dependencies across gait cycles using Transformer encoders, directly leading to a publication accepted in **IEEE JBHI (CCF-B)**.

Intelligent Cardiac MRI Analysis System via Deep Learning and Unsupervised Registration |*Project Lead*

Mar. 2024 — Aug. 2024

- Spearheaded research on myocardial segmentation and keypoint detection utilizing U-Net and other CNN architectures.
- Co-designed an unsupervised population registration network tailored for ventricular registration and cardiac motion tracking tasks.
- Secured a National Award in the 2024 Biomedical Engineering Innovation Competition and successfully filed a **Software Copyright**.

Research Internship at OMRI Brain-Computer Interface Lab, SJTU |*Early Undergraduate Research*

Sep. 2023 — Mar. 2024

- Participated in brain region segmentation projects utilizing multimodal medical imaging data.
- Explored foundational technologies for 3D digital twin brain visualization and neurosurgery simulation.

Personal Summary

- Professional working proficiency in English (IELTS 7.0).
- **Academic & Career Vision**: Incoming Master's student at CUHK. Deeply committed to the frontiers of Multimodal Large Language Models (MLLM) and Embodied AI. Dedicated to **Human-centered Artificial Intelligence**, with a vision to inject interpretable generative policies into the physical world to better serve communities and industries.
- **Technical Philosophy**: As a geek-minded algorithm research intern, I hold absolute faith and enthusiasm for the realization of AGI. Convinced that the ultimate goal of intelligence is to empower humanity, I maintain a sharp engineering intuition and a strong intrinsic motivation for continuous learning amidst the fast-iterating AI waves.
- **Personal Life**: Passionate about Citywalk and hiking, grounding my thoughts on how AI can serve the community through real-world physical interactions. I sharpen my logical reasoning and global planning skills through playing Go (Weiqi), and occasionally draw inspiration for constructing digital worlds from the open environment of Genshin Impact.

张岑蔚

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教育经历

香港中文大学 | 计算机科学与技术 (信息工程类)| 硕士研究生 2026.09(预计入学)

2028 年应届生。

上海交通大学 | 生物医学工程 (电子信息类)| 工学学士 2022.09—2026.06(预计)

GPA: 3.63/4.3，获学业奖学金多次，全国大学生生物医学工程创新创业国家级竞赛二等奖。

本科毕设方向：**面向医学图像数据的 SAM 分层分布保持剪枝框架，结合几何感知结构剪枝与渐进式细粒度稀疏化，探索医学基础分割模型的高效部署。**

技术能力

- **语言**: 常用 Python,C++；熟悉深度学习框架 Pytorch。
- **工作流**: Linux, Shell, Git, GitHub。
- **技术工程底座**: 熟悉 **Generative Modeling** (Flow Matching, Diffusion, AR) 理论及流形最优传输机制；熟悉 **Transformer, DiT, ViT** 骨干网络架构；熟悉基础 LLM 框架 (基于 nanoGPT 复现和微调)；了解 VLA 等 Embodied AI 知识 (pi0，RDT 等)
- **其他**: 有容器化技术的实践经验，熟悉 docker 的使用。

工作经历

DolphinAI **海豚智声医疗科技有限公司** | 算法研究实习生 / 医学图像生成 2025.04—2025.12

- **生成式模型预研与验证**: 独立主导基于 **Diffusion Models** 的肝脏超声及多模态医学图像生成项目，针对医学影像高频噪声与细粒度特征，完成从算法选型到生成质量可行性的深度验证。
- **算法工程化与容器化部署**: 摒弃纯理论实验，深度参与模型落地的工程化改造。引入并熟练运用 **Docker** 容器化技术，搭建标准化的模型训练与推理环境，解决跨平台环境依赖隐患。
- **合成数据管道构建**: 探索并沉淀了医学场景下的 Synthetic Data 生产流程，为下游疾病诊断模型的数据增强提供了可靠的算法基线与工程流水线经验。

科研论文

Axiomatic On-Manifold Shapley via Optimal Generative Flows |***CCFA-ICML*** 审稿中

Cenwei Zhang*, *Lin Zhu**, *Manxi Lin, Lei You*†

- **生成流与最优传输底座**: 构建基于最优生成流的连续几何框架，利用动能最小化的 Wasserstein-2 测地线定义高维数据流形上的精确传输路径，有效解决复杂流形上的路径歧义与离散化漂移问题。
- **公理化模型归因体系**: 推导并确立梯度线积分作为满足几何效率公理的唯一泛函，从理论基底上解决深度生成模型及复杂神经网络解释中的“离流形幻觉（Off-manifold artifacts)”与基线选择随意性。
- **量化评估与理论保障**: 提出流一致性误差等对齐评估指标，在多个生成任务基准中超越现有 SOTA 方法；提供可证明的数学稳定性界限以应对 ODE/SDE 近似误差。

GraphGaitNet: Anatomically-Informed Graph Neural Networks on Ground Reaction Force for Parkinson’s Disease Diagnosis |***CCFB-JBHI*** *Accept*

Wenhao Li, ***Cenwei Zhang****, Shi Chang, Jingtong Zhao, Guanning Lin*†

- 参与 GraphGaitNet 模型框架研究，基于足底解剖结构与受力传递规律构建 GRF 图表示，并结合图结构刻画局部与全局空间依赖，用于帕金森病步态诊断。
- 参与多层残差图卷积与 Transformer 时序编码模块的实验研究，在 PhysioNet Parkinson’s gait 数据集上验证模型性能。论文报告模型准确率达到 **96.8%**。

项目经历

基于 nanoGPT 的 LLM 模型预训练微调 | 独立项目与实现 2026.01—2026.03

- **模型架构复现**: 基于 PyTorch 复现 GPT2 的训练与推理流程，完成多头自注意力、因果掩码（Causal Mask）与自回归生成（AR）等核心模块实现。
- **架构与算子优化**: 在基础框架上引入 RoPE、RMSNorm 等主流 LLM 架构设计，并集成 FlashAttention 以优化长序列建模中的显存占用与计算效率。
- **预训练管线搭建**: 在单张 H100 80G GPU 上构建完整的预训练与验证流程，采用 `bfloat16` 混合精度与梯度累积等策略，提升训练吞吐与硬件利用效率。

基于图神经网络与时序建模的步态分析系统 | *研究协作* 2024.10—2025.04

- 基于 GRF 图结构与解剖学先验，参与步态分析联合模型 GraphGaitNet 的设计与实现。
- 参与多层残差图卷积模块与时序建模模块的实现与实验。
- 结合 Transformer 编码器建模步态周期中的时序依赖关系，相关论文已投稿至 **JBHI(CCFB)**。

基于深度学习与无监督配准的心脏 MRI 智能分析系统 | *项目组长* 2024.03—2024.08

- 基于 U-Net 等 CNN 模型开展心肌分割与关键点检测相关研究。
- 参与设计无监督群体配准网络，用于心室配准与心脏运动跟踪任务。
- 项目获 2024 年生物医学工程创新竞赛国家级奖项，并申请**软件著作权**。

上海交通大学 OMRI 脑机接口课题组科研实习 | *本科初段科研实习* 2023.09—2024.03

- 参与多模态医学图像脑区分割项目。
- 参与三维数字孪生脑可视化与脑外科手术模拟相关探索。

个人总结

- 可以使用英语进行工作交流（IELTS 7.0）。
- **学术与职业愿景**：即将赴香港中文大学攻读硕士。对多模态大模型 (MLLM) 与具身智能 (Embodied AI) 等前沿领域充满信心。致力于 **Human-centered Artificial Intelligence**，期许将可解释的生成式策略注入物理世界并更好服务于社区和产业。
- **技术信仰**：作为一名极客派的算法研究实习生，对 AGI 的实现抱有绝对的信念与热忱。我坚信智能的终局是服务人类，习惯在快速迭代的 AI 技术浪潮中保持敏锐的工程直觉与持续学习的内驱力。
- **生活触角**：热爱 Citywalk 与徒步，在真实的物理空间中感知思考 AI 如何服务社区；喜欢在围棋的对弈中淬炼逻辑推演与全局规划能力，偶尔也在 Genshin Impact 的开放世界里寻找构建数字世界的灵感。